Linkmodel: sn (EXPERIMENTAL)

Parametrization

This is the link that map $p \in (0,1)$ into $x \in \Re$, where

$$F_a(x) = p$$

and F_a is the cummulative distribution function for the skew-normal distribution,

$$2\phi(x)\Phi(a^{1/3}x)$$

which is renormalized to have zero mean and unit variance (expression omitted).

Hyperparameters

The parameter a represented using the (standarized) skewness s (expression omitted) where

$$s = s_{\max}\left(2\frac{\exp(\theta)}{1+\exp(\theta)} - 1\right)$$

$s_{\max} = 0.988$, and the prior is defined on θ. There is a PC prior available for θ. The PC-prior is corrected for this bound, whereas the pc-prior in the R-functions `inla.pc.{r,p,q,d}sn` does not define a such bound (but use the bound defined by the Skew-Normal itself, 0.99527...).

Specification

Use `model="sn"` within `control.link`.

Hyperparameter spesification and default values

doc `Skew-normal link`

hyper

theta1

hyperid `49031`
name `skewness`
short.name `skew`
initial `0.00123456789`
fixed `FALSE`
prior `pc.sn`
param `10`
to.theta `function(x, skew.max = 0.988) log((1 + x / skew.max) / (1 - x / skew.max))`
from.theta `function(x, skew.max = 0.988) skew.max * (2 * exp(x) / (1 + exp(x)) - 1)`

theta2

hyperid `49032`
name `intercept`
short.name `p0`
initial `0`
fixed `FALSE`
prior `linksnintercept`

param `0 0`
to.theta `function(x) log(x / (1 - x))`
from.theta `function(x) exp(x) / (1 + exp(x))`

pdf `linksn`

Example

Notes

- The link-function is also available as R-functions `inla.link.sn` and `inla.link.invsn`
- This link-model is experimental for the moment.
- Setting the initial value for the hyperparameter "intercept" to infinity, will remove the intercept from the link-model.